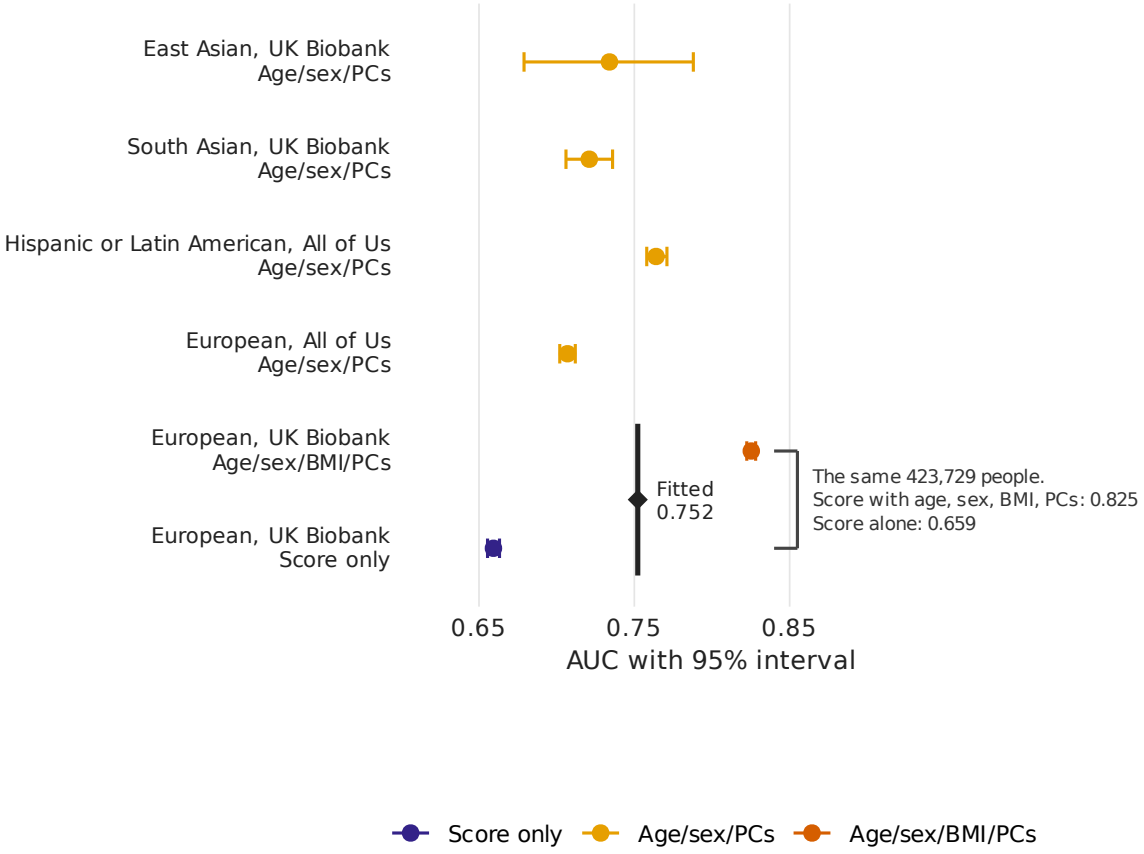
Why the weighted lm has σ = 11.5: one score evaluated twice on the same people

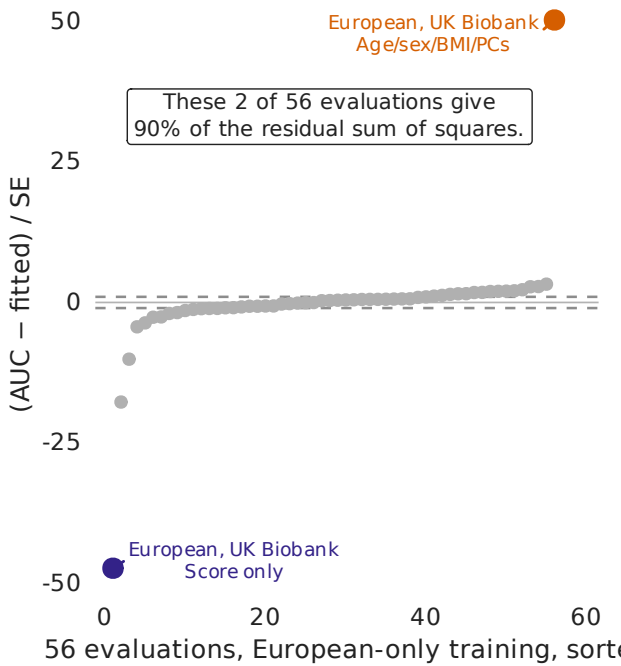
A. The six evaluations of PGS000805

Two are Europeans in UK Biobank, once with BMI in the model and once without. The weighted lm can fit only one value for both.



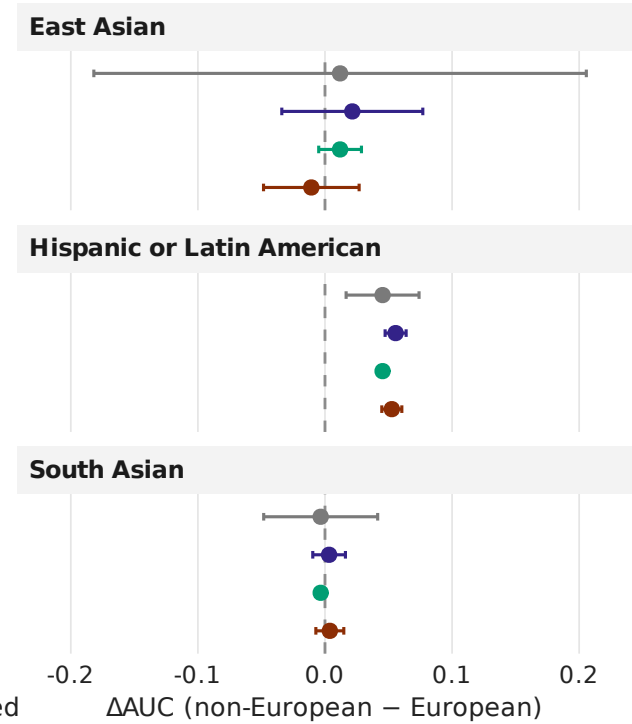
B. Where σ = 11.5 comes from

Distance of each evaluation from its fitted value, in standard errors. If the standard errors were right, most would fall between the dashed lines.



C. The ancestry differences

Coefficient and 95% interval from each weighted lm, with the covariance model for reference. Grey and green share the point; only σ differs.



Weighted lm as fitted (σ = 11.5)
Without the BMI evaluation (σ = 3.3)
Standard errors taken as known (σ = 1)
Covariance model (ρ = 0.6)

Type 2 diabetes, European-only training. In lm, the weights $1/SE^2$ only say how much each evaluation counts relative to the others. lm then estimates its own error scale σ and multiplies every standard error by it. Removing the BMI evaluation lowers σ from 11.5 to 3.3. Even then, the weighted lm treats evaluations on the same people as independent: this figure is about the weights, not about shared samples.

PGS000804 (multi-ancestry training) has the same pattern on African, European, and Hispanic or Latin American sample sets. The main ΔAUCs are not affected, because Stage 2 pairs European and non-European evaluations only within one covariate set.